

## Sequences and series



- 1 Describe the difference between a sequence and a series.
- 2 Find the third term in the sequence  $4, -5, 6, -7, 8$ .
- 3 Find the next number in the sequence  $16, 36, 64, 100, 144, \dots$
- 4 Consider the sequence  $a_n$ . If the first term is  $a_1$ , write the notation used for the fifth term of the sequence.
- 5 Determine whether each sequence is finite or infinite:
  - a  $1, 2, 3, 4, 5, 6$
  - b  $-1, -2, -3, -4, -5, -6, \dots$
  - c  $-1, -2, -3, -4, -5$
  - d  $1, 2, 3, 4, 5, 6, \dots$
- 6 If all the dates in July were used to form a sequence, would the sequence be finite or infinite?
- 7 Identify the following sequences as finite or infinite:
  - a  $a_1 = 2; a_n = 2a_{n-1}$  for  $n \geq 2$
  - b  $a_2 = 3; a_n = a_{n-1} + a_{n-2}$  for  $3 \leq n \leq 9$
- 8 State the first five terms of the following sequences. Round terms to four decimal places where necessary.
  - a  $a_n = 3n - 3$
  - b  $a_n = -2^n$
  - c  $a_n = 2^{n+1}$
  - d  $a_n = (-2)^n (3n)$
  - e  $a_n = \frac{1}{\sqrt{n}}$
  - f  $a_n = \left(1 + \frac{1}{n}\right)^n$
- 9 State the first four terms, in exact form, of the following sequences:
  - a  $a_n = \frac{n+4}{n+3}$
  - b  $a_n = \frac{5n-1}{n^2+5}$
  - c  $a_n = \left(\frac{1}{2}\right)^n (n-3)$
  - d  $a_n = (-1)^{n-3} (n-1)$

10 For each of the following sequences, state the first four terms:



- a  $a_1 = 4, a_n = na_{n-1} - 5$  for  $n > 1$
- b  $a_1 = 3, a_n = a_{n-1} + n$  for  $n > 1$
- c  $a_1 = -2, a_n = 2n + 3a_{n-1}$  for  $n > 1$
- d  $a_1 = 3, a_2 = 5, a_n = a_{n-1} \times a_{n-2}$  for  $n > 2$
- e  $a_1 = 3, a_2 = 2, a_n = 3(a_{n-1})^2 - 4a_{n-2}$  for  $n > 2$

## Sigma notation

11 Write the following series using sigma notation:

- a  $4 + 8 + 12 + 16 + 20 + \dots$
- b  $-3 + 9 - 27 + 81 - 243 + 729$
- c  $\frac{1}{1^3} + \frac{1}{2^3} + \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3}$
- d  $\frac{1}{3 \times 1} + \frac{1}{3 \times 2} + \frac{1}{3 \times 3} + \dots + \frac{1}{3 \times 7}$
- e  $\frac{2}{5+1} + \frac{2}{5+2} + \frac{2}{5+3} + \dots + \frac{2}{5+11}$
- f  $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots + \frac{1}{1024}$

12 Find the value of the following:

- a  $\sum_{r=1}^5 (4r - 4)$
- b  $\sum_{k=3}^5 \frac{1}{k}$
- c  $\sum_{r=1}^4 \frac{1}{r+2}$
- d  $\sum_{k=1}^5 (-1)^{k-1} k^2$
- e  $\sum_{r=-2}^1 (3r^2 - 1)$
- f  $\sum_{n=1}^5 5(3)^n$
- g  $\sum_{r=1}^5 (3^r + 1)$
- h  $\sum_{k=0}^3 ((-3)^k + 1)$
- i  $\sum_{n=1}^5 n^{n-3}$
- j  $\sum_{n=-3}^0 (n^2 - 4n)$

13 Given that  $x_1 = 2, x_2 = 3, x_3 = 4, x_4 = 5$  and  $x_5 = 6$ , find the value of  $\sum_{n=1}^5 x_n$ .

14 Given that  $x_1 = 1, x_2 = 0, x_3 = -1, x_4 = -2$  and  $x_5 = -3$ , find the value of  $\sum_{j=1}^5 2x_j$ .

15 Given that  $x_1 = -1, x_2 = 0, x_3 = 1, x_4 = 2$  and  $x_5 = 3$ , find the value of  $\sum_{r=1}^5 (4x_r + 2)$ .



17 Given that  $x_1 = 2$ ,  $x_2 = 1$ ,  $x_3 = 0$ ,  $x_4 = -1$  and  $x_5 = -2$ , find the value of the following:

a  $\sum_{i=1}^4 (x_i^2 - 3)$

b  $\sum_{n=2}^4 \left( \frac{x_n - 4}{x_n + 2} \right)$

18 Consider  $f(x) = 4x + 1$ ,  $x_1 = -1$ ,  $x_2 = -2$ ,  $x_3 = -3$ ,  $x_4 = -4$  and  $\delta x = \frac{3}{2}$ .

a  $\sum_{k=1}^4 f(x_k) \delta x$

b  $\sum_{x=1}^5 f(x)$

19 Consider the series  $\sum_{i=1}^7 (i + i^3)$  and  $\sum_{i=1}^7 i + \sum_{i=1}^7 i^3$ .

a Write each series in expanded form.

b Do the two sums have the same value?

c Is  $\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$ ?

## Applications

20 A doctor administered pain medication in the following amounts in milligrams over four days after their patient underwent surgery:

| Year ( $n$ )   | 1    | 2    | 3      | 4     |
|----------------|------|------|--------|-------|
| Dosage ( $m$ ) | 2000 | 1520 | 1155.2 | 878.0 |

Find the recursive formula which models this sequence. Round the values to two decimal places if necessary.

21 Christa deposits \$1 into a new bank account while Jenny deposits \$38 into a new bank account. The next day, Christa adds \$2 to the account, and doubles how much she adds each day thereafter. Jenny adds \$5 each day.

a List the amount Christa had in her account each day for the first four days.

b List the amount Jenny had in her account each day for the first four days.

c On which day will they have the same amount of money in their bank accounts?